

Day 34 - Dynamic Follow-Up and Adaptive Interview Engine

1. Objective

Day 34 focused on extending the AI interview system with a dynamic follow-up engine.

The engine analyzes candidate responses and determines whether an additional question is required based on response quality, length, vagueness, and confidence indicators.

2. Dynamic Follow-Up Engine

The new component is:

`interview_ai/follow_up_engine.py`

The main class is:

`DynamicFollowUpEngine`

Responsibilities

- Detect empty responses
- Detect vague responses
- Detect very short responses
- Trigger clarification questions
- Trigger deeper follow-up questions
- Trigger example-based questions
- Trigger scenario-based questions
- Identify sufficient responses
- Limit follow-up attempts
- Prevent repetitive follow-up questions
- Track follow-up conversation state
- Reset the follow-up state when required

3. Response Analysis

The engine classifies candidate responses into several conditions.

Empty Response

Action:

clarification

Example:

"Could you please provide a little more detail?"

Vague Response

Examples include:

- Yes
- No
- Okay
- Fine
- Maybe
- Not sure

Action:

clarification

Very Short Response

Responses containing two words or fewer trigger clarification.

Short Response

Responses containing up to five words trigger a deeper follow-up.

Action:

deepening

Confident Response

Responses containing confidence-related markers such as:

- led
- developed
- implemented
- designed
- managed
- built
- deployed
- improved

can trigger a scenario-based follow-up.

Action:

scenario

Detailed Response

Longer responses containing sufficient contextual information can trigger an example-based follow-up.

Action:

example

Sufficient Response

Responses that provide enough information without requiring additional probing are allowed to continue without a follow-up.

4. Follow-Up Types

The engine supports four follow-up types:

1. Clarification
2. Deepening
3. Example
4. Scenario

Each follow-up contains:

- Follow-up type
- Related question ID
- Generated follow-up text

5. Follow-Up Limit

The engine uses a maximum follow-up limit of:

2

After the maximum number of follow-ups has been reached, additional follow-ups are disabled.

The engine records:

- Follow-up count
- Eligibility
- Last trigger
- Previously asked follow-up questions

6. Duplicate Prevention

The engine stores previously asked follow-up questions.

If the same follow-up text has already been used, it is not generated again.

This prevents repetitive questioning during an interview.

7. Interview State

The engine maintains a state containing:

- follow_up_count
- follow_up_eligible
- last_trigger
- asked_follow_ups

The state can be retrieved using:

```
get_state()
```

The state can be reset using:

```
reset()
```

8. Existing Interview Engine Compatibility

The Day 34 implementation was tested together with the existing Day 33 Interview Engine.

This ensured that the new adaptive follow-up functionality did not break the existing interview functionality.

9. Testing

Day 34 Follow-Up Engine Tests

Result:

11 passed

Day 33 + Day 34 Interview Regression

Result:

24 passed

The regression confirms that the existing Day 33 interview engine and the new Day 34 follow-up engine are working together successfully.

10. Day 34 Outcome

Day 34 successfully introduced dynamic follow-up and adaptive questioning capability into the interview system.

The system can now analyze candidate responses and decide whether to:

- Continue the interview
- Request clarification
- Ask for more detail
- Request an example
- Ask a scenario-based question

The implementation is covered by automated tests and has passed the Day 33 regression suite.